STAFF REVIEW · H446 035 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

Do Now — source answers

1. An empty string is invalid because it contains no characters. The program should reject it, display an appropriate message and not store the username.

2. 1 + 0 + 1 = 2 = 10_2, so Sum = 0 and CarryOut = 1.

3. 12. After each iteration the totals are 2, 6 and 12.

4. (0,1) → Sum 1, Carry 0; (1,0) → Sum 1, Carry 0; (1,1) → Sum 0, Carry 1. In the last case 1 + 1 = 10_2, so the sum bit is 0 and the carry bit is 1.

I Do — source worked steps

1. Direction — CODE:
right shift

2. Amount — CODE:
two places

3. Final answer — CODE:
Shift right by two places.

We Do — guided CHECK

1. Complete both parts
CODE:
result = 00110000
effect = multiply by 4

2. Improve the answer
CODE:
"It is quicker."
CODE:
A BST can reduce the remaining search space after each comparison, whereas an unsorted array may require a linear search.

3. Complete the same expression
CODE:
¬(¬A ∧ ¬B) = ?
CODE:
¬(¬A ∧ ¬B) = A ∨ B

4. Improve the answer
CODE:
It adds ...
CODE:

It adds A, B and Cin. S is the result bit and Cout is the carry-out.

You Do — source mark schemes

1. Data Types — mark, fix, retry
CODE:
Mark the Teams OCR questions immediately.

For every lost mark:
1. identify the exact step or fact missing
2. write the correct rule
3. close the mark scheme
4. retry the question

Do not finish this block with an uncorrected error.

2. Data Structures — mark, fix, retry
CODE:
Mark the Teams OCR questions immediately.

Check especially:
• pointer/index language
• LIFO vs FIFO
• traversal/search explanations
• BST ordering
• hash-table lookup/collisions

Correct the reason for the error, then retry.

3. Boolean Algebra — mark, fix, retry
CODE:
Mark the Teams OCR questions immediately.

Check especially:
• Boolean symbols and brackets
• one algebra law per transformation
• valid K-map groups
• D-type clock behaviour
• Sum/Carry and Cin/Cout

Retry any incorrect question without looking.

4. Personal checklist
CODE:
Write two exact checks for the test.

Examples:
• "Check I have treated the exponent as signed."
• "Check pointer fields contain indexes, not data."

• "Check K-map groups are powers of two."
• "Check every truth-table input combination is present."

Use your own two weakest-error patterns.

Exit Ticket — source answer

1. [No safely extracted source model: staff must check]